

An abstract, complex wireframe structure composed of numerous interconnected lines and polygons, resembling a stylized, multi-faceted object or a network diagram. The structure is rendered in a light gray color against a dark gray background.

Generative AI

Which face is real?



A



B



C

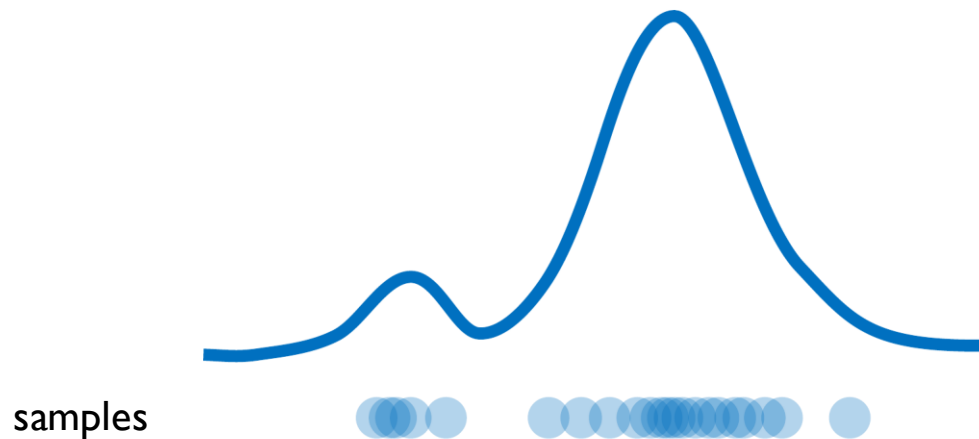
Generative modeling

Goal: Take as input training samples from some distribution and learn a model that represents that distribution

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Density Estimation

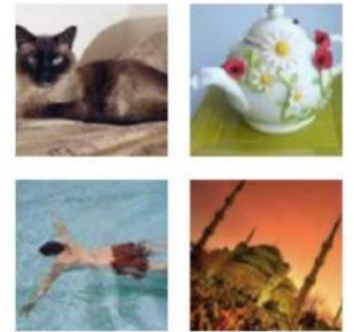


Sample Generation



Input samples

Training data $\sim P_{data}(x)$



Generated samples

Generated $\sim P_{model}(x)$

How can we learn $P_{model}(x)$ similar to $P_{data}(x)$?

Why generative models? Debiasing

Capable of uncovering **underlying latent variables** in a dataset



Homogeneous skin color, pose

VS



Diverse skin color, pose, illumination

How can we use latent distributions to create fair and representative datasets?

Why generative models? Outlier detection

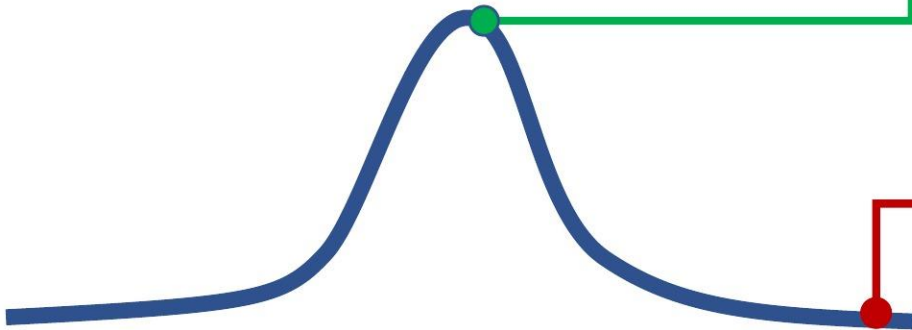
- **Problem:** How can we detect when we encounter something new or rare?
- **Strategy:** Leverage generative models, detect outliers in the distribution
- Use outliers during training to improve even more!

95% of Driving Data:

(1) sunny, (2) highway, (3) straight road



Detect outliers to avoid unpredictable behavior when training



Edge Cases



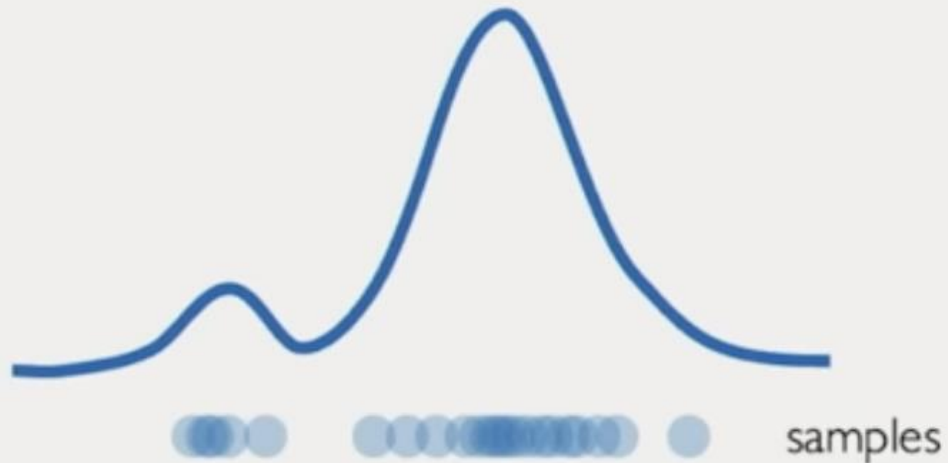
Harsh Weather



Pedestrians

Why generative models? Sample generation

- **Generative models learn probability distributions**
- **Sampling** from that distribution → new data instances
- **Backbone of Generative AI:** generate language, images, and more



Applications of Generative AI



Text Generation:

Content creation
Language translation
Summarization



Image Generation:

Art and design
Medical imaging
Video game development



Music Generation:

Composition
Sound effects



Drug Discovery:

Designing new molecules

Image generation



Text-to-Music (T2M) generation



(Stability AI) Evans, Zach, Julian D. Parker, C. J. Carr, Zack Zukowski, Josiah Taylor, and Jordi Pons. ["Long-form music generation with latent diffusion."](#) arXiv 2024.

Text-to-Video (T2V) generation



#Sora (OpenAI) Tim Brooks, Bill Peebles, Connor Holmes, Will DePue, Yufei Guo, Li Jing, David Schnurr, Joe Taylor, Troy Luhman, Eric Luhman, Clarence Wing Yin Ng, Ricky Wang, Aditya Ramesh. [Video generation models as world simulators](#). OpenAI 2024.

Course Outline

Historical Context and Evolution

- **2014:** Ian Goodfellow introduced GANs, allowing machines to generate realistic images by pitting two networks against each other.
- **2017:** Transformers emerged with the Attention is All You Need paper, laying the foundation for models like BERT and GPT.
- **2020:** OpenAI's GPT-3 stunned the world with its ability to generate coherent, context-aware text.
- **2022:** Diffusion models gained attention with tools like DALL-E and Stable Diffusion, redefining creativity.
- **2023:** GPT-4 and Gemini → multimodal AI; LLaMA → open-source boom.
- **2024:** Rise of reasoning LLMs (Claude 3, Mistral), Agentic AI frameworks (LangChain, AutoGPT), MoE architectures.
- **2025-2026:** Focus on safe, reliable LLMs → hallucination reduction, reasoning (CoT, ToT), and agentic RAG.

Note: VAEs (2013) and diffusion's original paper (2015). Good ideas often wait years for the compute to catch up.

Challenges in GenAI

- Ethical Concerns: Bias, deepfakes
- Hallucinations in AI-generated content
- High computational costs (resource intensive)
- Data privacy concerns

Discriminative Vs Generative Model



Machine learning models can be broadly categorized into discriminative and generative models.



Each type of model has different goals, strengths, and applications.

Discriminative Models

- Discriminative models focus on distinguishing between different classes in the data.
- They model the decision boundary between classes.
- Given input data x , predict the label y .
- Learns the $P(y|x)$
- **Examples:**
 - Logistic Regression
 - Neural Networks
 - Convolutional Neural Networks
- **Advantages:**
 - Often achieve higher accuracy on classification tasks.
 - Directly optimize the decision boundary.

Generative Models

- Generative models focus on modeling the distribution of the data.
- They learn to generate new data points that resemble the training data.
- Model the probability distribution $P(x)$.
- If dataset is labelled, estimates/learn the $P(x|y)$ - data conditioned on specific labels
- **Examples:**
 - Gaussian Mixture Models (GMMs)
 - Hidden Markov Models (HMMs)
 - Variational Autoencoders (VAEs)
 - Generative Adversarial Networks (GANs)
- **Advantages:**
 - Can generate new data samples.
 - Useful for unsupervised learning tasks.

Key Differences

Discriminative

Learn the Decision Boundary

Model $P(y|x)$.

Requires Labelled Data

Generative

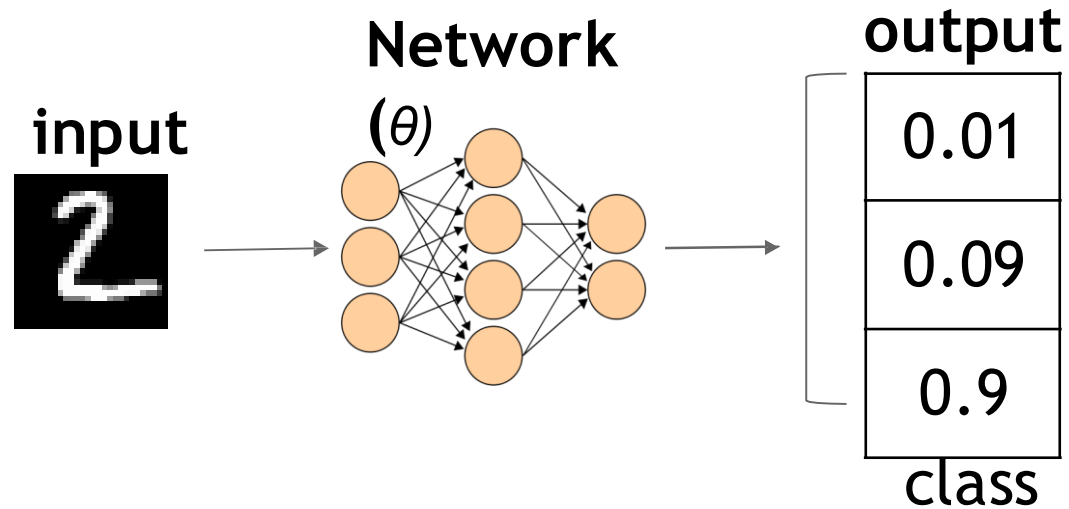
Learn the distribution of the data.

Model $P(x)$.

Can work with unlabeled data.

$P_{\theta}(Y | X)$: Discriminative Models

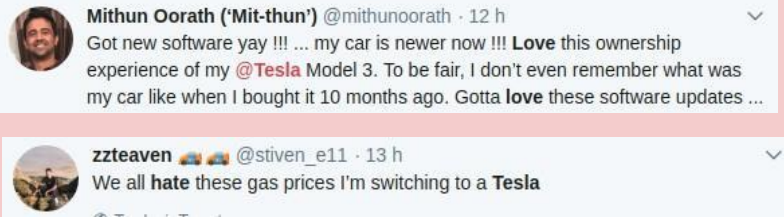
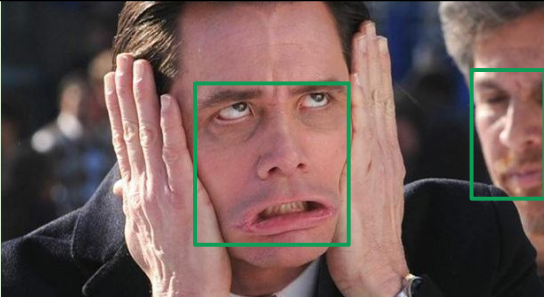
Discriminative model: Tell me the probability of some 'Y' responses given 'X' input.



$$P_{\theta}(Y | X = [\text{pixel}_1, \text{pixel}_2, \dots, \text{pixel}_{784}])$$

$P_{\theta}(Y|X)$: Discriminative Models

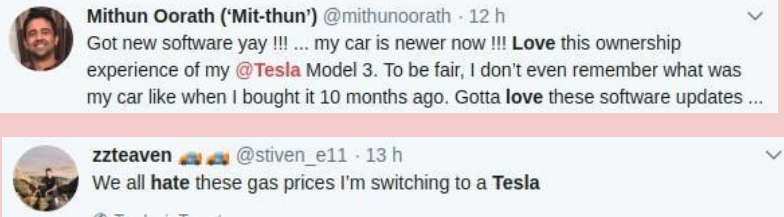
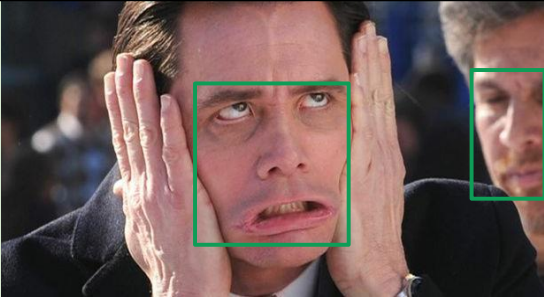
Slide credit:
Albert Pumarola (UPC 2019)

	Classification	Regression
Text		Prob. of being a Potential Customer
Image	 Jim Carrey	
Audio	 What Language?	Speech Translation

X=Data
Y=Labels
 θ = Model parameters

Discriminative Modeling
 $P_{\theta}(Y|X)$

$P_{\theta}(X)$: Generative Models

	Classification	Regression	Generative
Text		Prob. of being a Potential Customer	"What about Ron magic?" offered Ron. To Harry, Ron was loud, slow and soft bird. Harry did not like to think about birds.
Image	 Jim Carrey		
Audio	 What Language?	Language Translation	Music Composer and Interpreter MuseNet Sample

X=Data
Y=Labels
 θ = Model parameters

Discriminative Modeling
 $P_{\theta}(Y|X)$

Generative Modeling
 $P_{\theta}(X)$

Two ways of studying the same letter

DISCRIMINATIVE

“Which bin does this go in?”

A mail sorter doesn't need to know how each sender writes — only enough about the envelope to decide which bin it belongs in. It learns the boundary between categories directly.

models $P(y | x)$

GENERATIVE

“What does each sender's writing look like?”

A forger studies a sender's handwriting so thoroughly that they can produce a brand-new letter as if that person wrote it. It learns the full shape of the data itself.

models $P(x, y)$

What each model actually learns

Discriminative

Learns the conditional distribution of the label given the input:

$$P(y | x)$$

- Directly maps input $\rightarrow$ label
- Optimizes for the boundary between classes
- Never models what x itself looks like
- Cannot sample new x values

Generative

Learns the full joint distribution of inputs and labels:

$$P(x, y) = P(x | y) P(y)$$

- Models what each class looks like
- Recovers $P(y | x)$ via Bayes' rule if needed
- Can compute how “likely” any x is
- Can sample brand-new x values

It isn't just math — it changes what the model can do

	DISCRIMINATIVE	GENERATIVE
Classify inputs	✓	✓
Generate brand-new data	—	✓
Estimate how “typical” an input is (for outlier / anomaly detection)	—	✓
Work naturally with missing features	—	✓
Tends to need less labeled data (can learn from unlabeled x via density estimation)	—	✓
Often reaches higher raw classification accuracy with abundant labeled data	✓	—

Discriminative models

Logistic Regression

Learns a linear boundary between classes

Support Vector Machines

Maximizes the margin between classes

Decision Trees / Random Forests

Splits input space into labeled regions

k-Nearest Neighbors

Labels x by its closest neighbors

Conditional Random Fields

Boundary-based labeling for sequences

CNN / Transformer classifiers

e.g. ResNet for images, BERT for sentiment

Generative models

Naive Bayes

Models $P(\text{word} \mid \text{class})$ for text classification

Gaussian Mixture Models

Represents data as a mixture of “blobs”

Hidden Markov Models

Models how sequences are generated over time

Variational Autoencoders

Learns a compressed, samplable data space

GANs & Diffusion Models

Generate realistic images (e.g. Stable Diffusion)

Autoregressive LMs (GPT)

Generates text one token at a time from $P(x)$

The comparison, in one table

Aspect	Discriminative	Generative
Models		
Learning goal		
Can generate new samples?		
Handles missing / unlabeled data		
Typical labeled-data needs		
Classic examples		

The comparison, in one table

Aspect	Discriminative	Generative
Models	$P(y \mid x)$	$P(x, y)$
Learning goal	The boundary between classes	The distribution of the data itself
Can generate new samples?	No	Yes
Handles missing / unlabeled data	Poorly	Naturally, by marginalizing
Typical labeled-data needs	Higher	Can be lower (uses density est.)
Classic examples	Logistic Regression, SVM, CNNs	Naive Bayes, GANs, Diffusion, GPT

Same task, two philosophies: spam filtering

Training data: 100 emails, 40% spam. The word “win” appears in 60% of spam emails but only 5% of real (“ham”) emails.

Naive Bayes (generative)

Learns $P(\text{word} \mid \text{spam})$, $P(\text{word} \mid \text{ham})$, and $P(\text{spam})$, then applies Bayes' rule:

$$\begin{aligned} P(\text{spam} \mid \text{“win”}) &\propto P(\text{“win”} \mid \text{spam}) \cdot P(\text{spam}) \\ &= 0.60 \times 0.40 = 0.24 \end{aligned}$$

$$\begin{aligned} P(\text{ham} \mid \text{“win”}) &\propto P(\text{“win”} \mid \text{ham}) \cdot P(\text{ham}) \\ &= 0.05 \times 0.60 = 0.03 \end{aligned}$$

Normalize $\rightarrow P(\text{spam} \mid \text{“win”}) \approx 0.89$. Bonus: it can also generate a fake spam-sounding email by sampling words from $P(\text{word} \mid \text{spam})$.

Logistic Regression (discriminative)

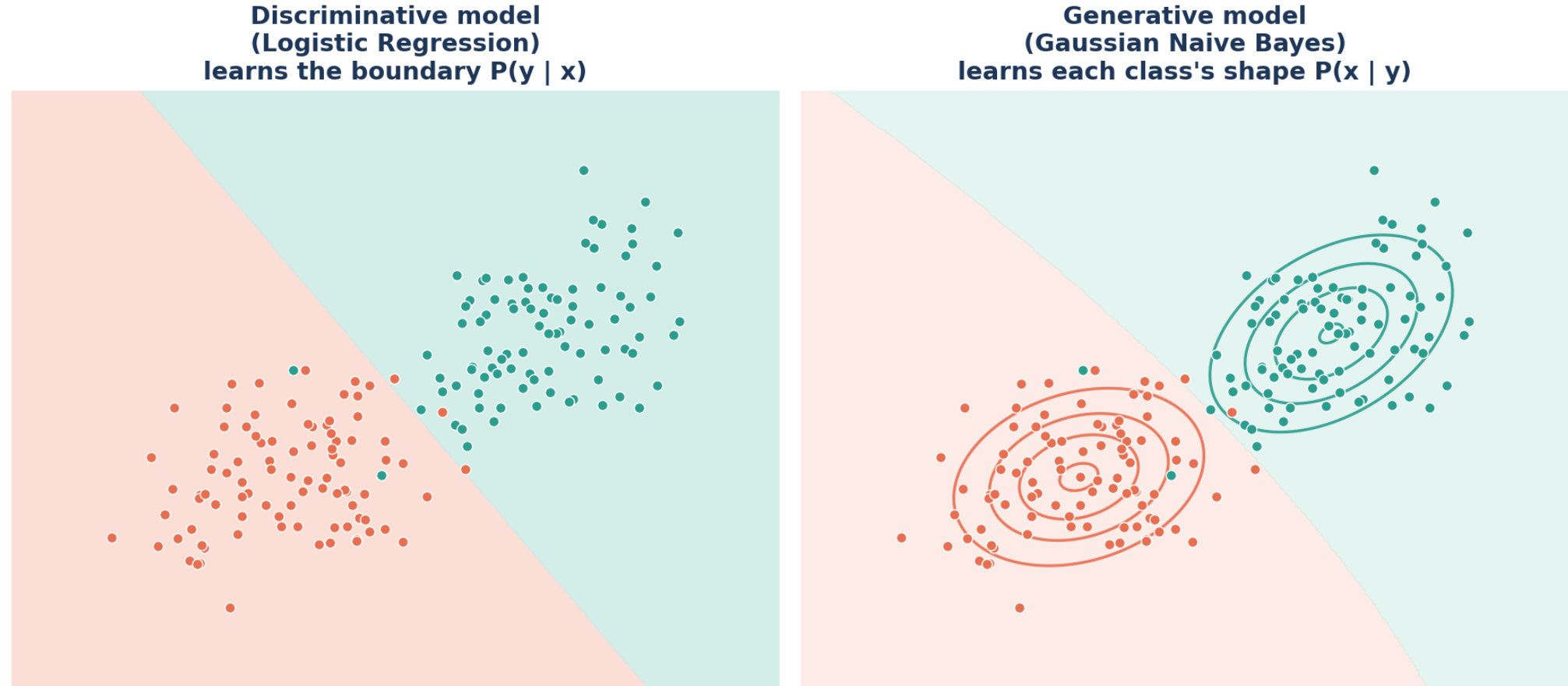
Learns weights that map word-features directly to a spam probability — it never asks what a typical spam email looks like:

$$P(\text{spam} \mid \mathbf{x}) = \sigma(\mathbf{w} \cdot \mathbf{x} + b)$$

Fit directly by maximizing classification accuracy on labeled emails — no intermediate model of “what spam looks like” is ever built.

It cannot write a new spam email — only score ones that already exist.

What each model actually learns from identical data



Both were trained on the same two clusters of points. The discriminative model only ever drew a line. The generative model described the shape of each class — the boundary is a side effect.

Only one of these can invent something new

**Discriminative model
can only label points that
already exist**



It has no notion of what an
“even more typical” point looks like.

**Generative model can sample
brand-new, never-seen points
from $P(x | y)$**



Stars = new synthetic examples drawn
from the model — not in the training set.

This is the ability that powers GPT (sampling the next token), GANs and diffusion models (sampling pixels) — all generative models at their core.

Mapping today's AI systems onto both families

Discriminative

- BERT / RoBERTa — sentiment, NER, intent classification
- ResNet / EfficientNet — image classification
- Fraud & spam filters, credit scoring models
- Most search / recommendation ranking models

Generative

- GPT-family models — generate text token by token
- Stable Diffusion / DALL·E — generate images from noise
- Voice synthesis (e.g. WaveNet-style models)
- Variational Autoencoders — compress and resynthesize data

The lines blur — and one model proves it

The Generative Adversarial Network (GAN)

A GAN is literally built from both halves of this lecture, competing with each other:

Generator

A generative model that learns $P(x)$ for realistic images and tries to fool the discriminator with fabricated samples.

Discriminator

A discriminative model that learns $P(\text{real} \mid x)$ and tries to catch the generator's fakes.

So which one should you reach for?

Choose Discriminative when...

Choose Generative when...

So which one should you reach for?

Choose Discriminative when...

- The goal is pure prediction accuracy
- You have plenty of labeled data
- You never need to generate or sample data
- Only the boundary between classes matters

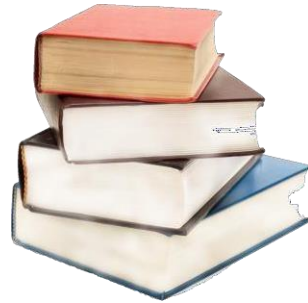
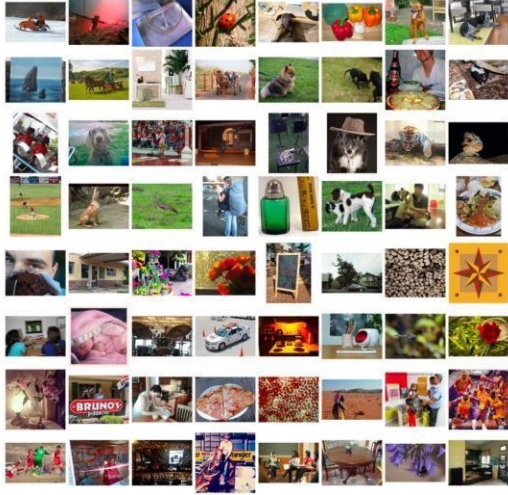
Choose Generative when...

- You need to generate new data (text, images, audio)
- Labeled data is scarce but unlabeled data is plentiful
- You need to detect outliers or handle missing features
- You want an interpretable model of the data itself

Key takeaways

- 1 Discriminative models learn $P(y \mid x)$: a direct map from input to label.
- 2 Generative models learn $P(x, y)$: the full distribution the data came from.
- 3 Only generative models can sample new, never-seen data.
- 4 Neither family is strictly “better” — the right choice depends on the task.
- 5 The distinction still organizes today's AI landscape: classifiers vs. GPT / diffusion models.

$P_{\theta}(X)$: Generative Models

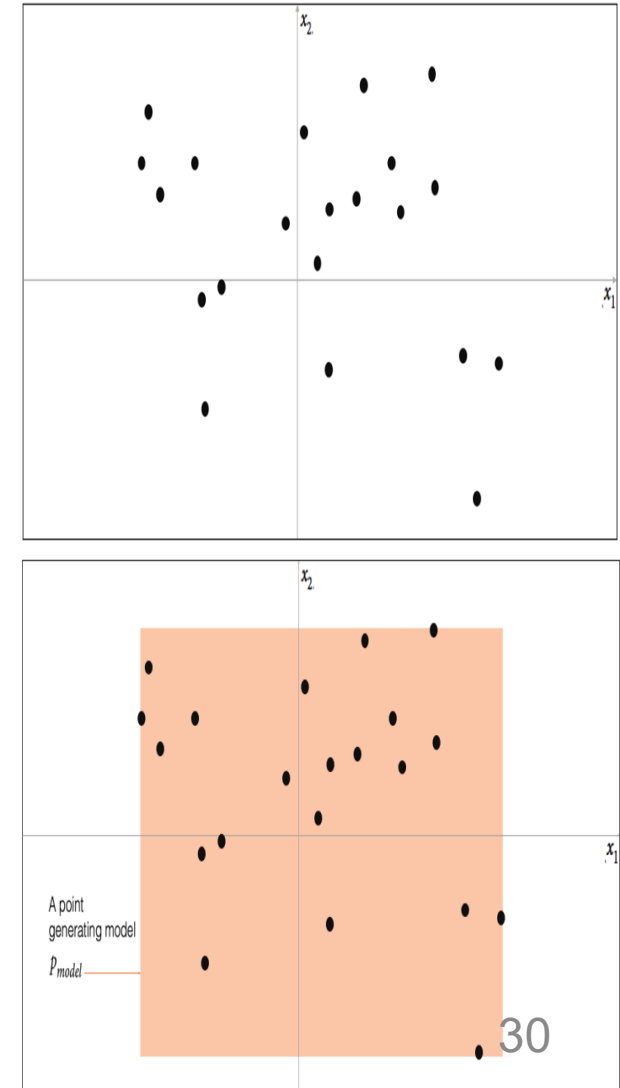


Each real sample x_i comes from an M -dimensional probability distribution $P(X)$.

$$X = \{x_1, x_2, \dots, x_N\}$$

Generative Modeling Framework

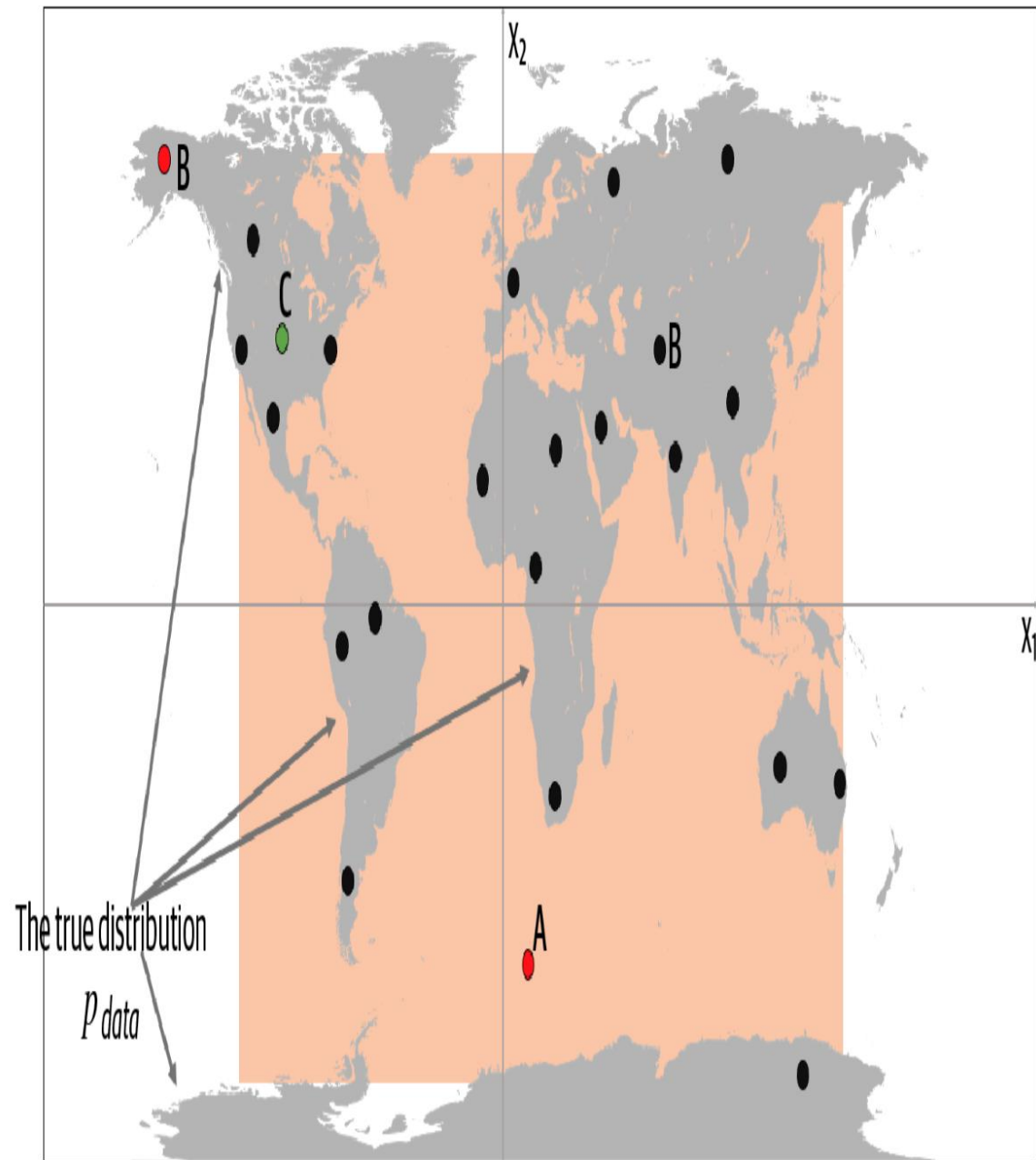
- A rule has generated some set of points
- Let's call the rule P_{data}
- Task is to select a point that has been generated by the same rule
- Try to estimate the rule
- Let's call the estimation P_{model}



Generative Modelling Framework

- We have a dataset of observations $\mathbf{X}$.
- We assume that the observations have been generated according to some unknown distribution, p_{data} .
- We want to build a generative model p_{model} that mimics p_{data} . If we achieve this goal, we can sample from p_{model} to generate observations that appear to have been drawn from p_{data} .
- The desirable properties of p are:
 - **Accuracy:** If $p_{model}(x)$ is high, x should look like it has been drawn from p_{data} . If $p_{model}(x)$ is low, x should *not* look like it has been drawn from p_{data} . The model should accurately capture the patterns and characteristics of the real data.
 - **Generation:** It should be possible to easily sample a new observation x from $p_{model}(x)$. If sampling is difficult or computationally expensive, the model is less practical.
 - **Representation:** It should be possible to understand how different high-level features (e.g. in our example, continents) are represented by p_{model} .

P_{data} Vs P_{model}



P_{data} Vs P_{model}

- **Point A** is an observation that is generated by our model but does not appear to have been generated by p_{data} as it's in the middle of the sea.
- **Point B** could never have been generated by p_{model} as it sits outside the orange box.
 - model has some gaps in its ability to produce observations across the entire range of potential possibilities.
- **Point C** is an observation that could be generated by p_{model} and also by p_{data}

